

```
import numpy as np
import pandas as pd
```

```
a1 = np.array([[[1,2,3],[6,9,0],[6,78,9]],[[4,5,6],[7,9,0],[8,9,7]],[[5,6,7],[8,9,7],[9,6,4]]])
a1
```

```
array([[[ 1,  2,  3],
         [ 6,  9,  0],
         [ 6, 78,  9]],

       [[ 4,  5,  6],
         [ 7,  9,  0],
         [ 8,  9,  7]],

       [[ 5,  6,  7],
         [ 8,  9,  7],
         [ 9,  6,  4]]])
```

```
a1[2][2][2]
```

```
np.int64(4)
```

```
a1.shape
```

```
(3, 3, 3)
```

```
a = np.array([12,3,7])
a
```

```
array([12,  3,  7])
```

```
a1.ndim
```

```
3
```

```
a1[:, :, ::1][:3][1:2]
```

```
array([[[4, 5, 6],
         [7, 9, 0],
         [8, 9, 7]]])
```

```
int(a1[2][2][2])
```

```
4
```

```
a1[1,0,[0,2]]
```

```
array([4, 6])
```

```
np.array([[1,2,3],[6,8,9]]).shape
```

```
(2, 3)
```

```
np.array([[[5,7,8],[9,7,5]]])
```

```
array([[5, 7, 8],
       [9, 7, 5]])
```

```
np.array([[[5,7,8],[9,7,5]],[[6,8,9],[8,9,4]],[[8,9,4],[89,90,8]]])
```

```
array([[[ 5,  7,  8],
         [ 9,  7,  5]],

       [[ 6,  8,  9],
         [ 8,  9,  4]],

       [[ 8,  9,  4],
         [89, 90,  8]]])
```

```
A1 = np.arange(1,11)
A1
```

```
array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10])
```

```
A1.reshape(2,5).ndim
```

```
2
```

```
np.arange(1,49).reshape(3,4,4)
```

```
array([[[ 1,  2,  3,  4],
         [ 5,  6,  7,  8],
         [ 9, 10, 11, 12],
         [13, 14, 15, 16]],

       [[17, 18, 19, 20],
         [21, 22, 23, 24],
         [25, 26, 27, 28],
         [29, 30, 31, 32]],

       [[33, 34, 35, 36],
         [37, 38, 39, 40],
         [41, 42, 43, 44],
         [45, 46, 47, 48]]])
```

```
np.arange(1,37).reshape(3,4,3).ndim
```

```
3
```

```
np.array([1,2,3]).ndim
```

```
1
```

```
np.array([[1,2,3],[8,9,0]])
```

```
array([[1, 2, 3],
       [8, 9, 0]])
```

```
np.array([[1,2,3],[8,9,0]]).ndim
```

```
2
```

```
np.array([[[3,6,8],[8,9,0]],[[7,8,5],[9,7,4]]])
```

```
array([[ [3, 6, 8],
        [8, 9, 0]],

       [[7, 8, 5],
        [9, 7, 4]]])
```

```
np.array([[[3,6,8],[8,9,0]],[[7,8,5],[9,7,4]],[[8,6,4],[1,7,4]]])
```

```
array([[ [3, 6, 8],
        [8, 9, 0]],

       [[7, 8, 5],
        [9, 7, 4]],

       [[8, 6, 4],
        [1, 7, 4]]])
```

```
np.array([[[3,6,8],[8,9,0]],[[7,8,5],[9,7,4]],[[8,6,4],[1,7,4]]]).ndim
```

```
3
```

```
np.array([[[3,6,8],[8,9,0]],[[7,8,5],[9,7,4]],[[8,6,4],[1,7,4]]]).shape
```

```
(3, 2, 3)
```

```
np.ones((3,6)).astype(np.int32)
```

```
array([[1, 1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1, 1]], dtype=int32)
```

```
np.zeros((3,6))
```

```
array([[0., 0., 0., 0., 0., 0.],
       [0., 0., 0., 0., 0., 0.],
       [0., 0., 0., 0., 0., 0.]])
```

```
np.random.random((5,6))*2
```

```
array([[0.97129226, 1.09543699, 0.68254631, 1.17222838, 0.18244244,
        1.46885153],
       [0.7368601 , 1.13583058, 0.95650777, 0.84000125, 0.33044211,
        0.52449343],
       [0.23194793, 1.45518085, 1.54047457, 1.61771041, 0.08660545,
        1.65750424],
       [0.66403813, 1.40417301, 0.70491838, 0.63302184, 0.4569225 ,
        0.15055646],
       [1.63446237, 0.85561802, 1.16211994, 1.76181607, 0.14324414,
        1.00168025]])
```

```
np.linspace(0,10,5).astype(np.int32)*2
```

```
array([ 0,  4, 10, 14, 20], dtype=int32)
```

```
a2 = np.arange(1,49).reshape(3,4,4)
a2
```

```
array([[[ 1,  2,  3,  4],
        [ 5,  6,  7,  8],
        [ 9, 10, 11, 12],
```

```
[13, 14, 15, 16]],  
  
[[17, 18, 19, 20],  
 [21, 22, 23, 24],  
 [25, 26, 27, 28],  
 [29, 30, 31, 32]],  
  
[[33, 34, 35, 36],  
 [37, 38, 39, 40],  
 [41, 42, 43, 44],  
 [45, 46, 47, 48]]])
```

```
a2 > 12 # relational operator
```

```
array([[False, False, False, False],  
       [False, False, False, False],  
       [False, False, False, False],  
       [ True,  True,  True,  True]],  
  
       [[ True,  True,  True,  True],  
        [ True,  True,  True,  True],  
        [ True,  True,  True,  True],  
        [ True,  True,  True,  True]],  
  
       [[ True,  True,  True,  True],  
        [ True,  True,  True,  True],  
        [ True,  True,  True,  True],  
        [ True,  True,  True,  True]])
```

```
d1 = np.arange(1,43).reshape(3,7,2)  
d1
```

```
array([[[ 1,  2],  
        [ 3,  4],  
        [ 5,  6],  
        [ 7,  8],  
        [ 9, 10],  
        [11, 12],  
        [13, 14]],  
  
       [[15, 16],  
        [17, 18],  
        [19, 20],  
        [21, 22],  
        [23, 24],  
        [25, 26],  
        [27, 28]],  
  
       [[29, 30],  
        [31, 32],  
        [33, 34],  
        [35, 36],  
        [37, 38],  
        [39, 40],  
        [41, 42]]])
```

```
d2 = np.array([[[1,2,4],[4,6,7]],[[4,6,7],[5,6,7]]])  
d2
```

```
array([[[1, 2, 4],  
        [4, 6, 7]],  
  
       [[4, 6, 7],  
        [5, 6, 7]]])
```

```
np.random.randint((2,3))
```

```
array([1, 2])
```

```
np.random.randint(1,10)
```

```
3
```

```
a = np.random.random((3,3))  
a
```

```
array([[0.32152097, 0.38801381, 0.13212286],  
       [0.82928784, 0.92372308, 0.83694277],  
       [0.25401518, 0.45208135, 0.04870296]])
```

```
a = np.round(a*100)  
a
```

```
array([[32., 39., 13.],  
       [83., 92., 84.],  
       [25., 45.,  5.]])
```

```
a = np.random.random((3,3))  
a = np.round(a*100)  
a
```

```
array([[ 34.,  79.,  40.],  
       [ 22.,   0., 100.],  
       [ 79.,  90.,  47.]])
```

```
np.max(a)
```

```
np.float64(100.0)
```

```
np.min(a)
```

```
np.float64(0.0)
```

```
np.sum(a)
```

```
np.float64(491.0)
```

```
np.prod(a)
```

```
np.float64(0.0)
```

```
np.max(a,axis = 1)
```

```
array([ 79., 100.,  90.])
```

```
np.max(a,axis = 0)
```

```
array([ 79.,  90., 100.])
```

```
np.sum(a,axis = 1)
```

```
array([153., 122., 216.])
```

```
np.sum(a,axis = 0)
```

```
array([135., 169., 187.])
```

```
np.prod(a,axis = 0)
```

```
array([ 59092.,      0., 188000.])
```

```
np.prod(a,axis = 1)
```

```
array([107440.,      0., 334170.])
```

```
np.mean(a)
```

```
np.float64(54.55555555555556)
```

```
np.mean(a,axis=1)
```

```
array([51.      , 40.66666667, 72.      ])
```

```
np.mean(a,axis=0)
```

```
array([45.      , 56.33333333, 62.33333333])
```

```
np.mod(a,3)
```

```
array([[1., 1., 1.],  
       [1., 0., 1.],  
       [1., 0., 2.]])
```

```
np.mod(a,5)
```

```
array([[4., 4., 0.],  
       [2., 0., 0.],  
       [4., 0., 2.]])
```

```
d2 = np.arange(12).reshape(3,4)
```

```
d3 = np.arange(12).reshape(4,3)
```

```
np.dot(d2,d3)
```

```
array([[ 42,  48,  54],  
       [114, 136, 158],  
       [186, 224, 262]])
```

```
np.dot(d2,d3).ndim
```

```
2
```

```
np.dot(d2,d3).shape
```

```
(3, 3)
```

```
np.round(np.mean(a))
```

```
np.float64(55.0)
```

```
np.floor(np.mean(a))
```

```
np.float64(54.0)
```

```
np.ceil(np.mean(a))
```

```
np.float64(55.0)
```

```
np.round(np.mean(a)*100)
```

```
np.float64(5456.0)
```

```
a1 = np.arange(10)
a2 = np.arange(12).reshape(3,4)
a3 = np.arange(8).reshape(2,2,2)
```

```
a1
```

```
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
a2
```

```
array([[ 0,  1,  2,  3],
       [ 4,  5,  6,  7],
       [ 8,  9, 10, 11]])
```

```
a3
```

```
array([[[0, 1],
        [2, 3]],

       [[4, 5],
        [6, 7]]])
```

```
a3[1,1,1]
```

```
np.int64(7)
```

```
a3[1,1,0]
```

```
np.int64(6)
```

```
a4 = np.arange(1,49).reshape(3,4,4)
a4
```

```
array([[[ 1,  2,  3,  4],
        [ 5,  6,  7,  8],
        [ 9, 10, 11, 12],
        [13, 14, 15, 16]],

       [[17, 18, 19, 20],
        [21, 22, 23, 24],
        [25, 26, 27, 28],
        [29, 30, 31, 32]],

       [[33, 34, 35, 36],
```

```
[37, 38, 39, 40],  
[41, 42, 43, 44],  
[45, 46, 47, 48]]])
```

```
a4[2,2,2:5]
```

```
array([43, 44])
```

```
a4[2,0:4,0:5]
```

```
array([[33, 34, 35, 36],  
       [37, 38, 39, 40],  
       [41, 42, 43, 44],  
       [45, 46, 47, 48]])
```

```
a5 = np.arange(1,13).reshape(3,4)  
a5
```

```
array([[ 1,  2,  3,  4],  
       [ 5,  6,  7,  8],  
       [ 9, 10, 11, 12]])
```

```
a5[2,:]
```

```
array([ 9, 10, 11, 12])
```

```
a5[:,1]
```

```
array([ 2,  6, 10])
```

```
a5
```

```
array([[ 1,  2,  3,  4],  
       [ 5,  6,  7,  8],  
       [ 9, 10, 11, 12]])
```

```
a5[1:3,1:3]
```

```
array([[ 6,  7],  
       [10, 11]])
```

```
a5[1:3,1:4]
```

```
array([[ 6,  7,  8],  
       [10, 11, 12]])
```

```
a5[0:2,2:4]
```

```
array([[3, 4],  
       [7, 8]])
```

```
a5[1:3,3:4]
```

```
array([[ 8],  
       [12]])
```

```
a6 = np.arange(1,49).reshape(3,4,4)  
a6
```



```
array([[ 1,  2,  3,  4],
       [ 5,  6,  7,  8],
       [ 9, 10, 11, 12],
       [13, 14, 15, 16]],

      [[17, 18, 19, 20],
       [21, 22, 23, 24],
       [25, 26, 27, 28],
       [29, 30, 31, 32]],

      [[33, 34, 35, 36],
       [37, 38, 39, 40],
       [41, 42, 43, 44],
       [45, 46, 47, 48]]])
```

```
a6[2,1:3,1:3]
```

```
array([[38, 39],
       [42, 43]])
```

```
a6[2,2:3,1:]
```

```
array([[42, 43, 44]])
```

```
a6[1:2,1:3,1:4]
```

```
array([[22, 23, 24],
       [26, 27, 28]])
```

```
a6
```

```
array([[ 1,  2,  3,  4],
       [ 5,  6,  7,  8],
       [ 9, 10, 11, 12],
       [13, 14, 15, 16]],

      [[17, 18, 19, 20],
       [21, 22, 23, 24],
       [25, 26, 27, 28],
       [29, 30, 31, 32]],

      [[33, 34, 35, 36],
       [37, 38, 39, 40],
       [41, 42, 43, 44],
       [45, 46, 47, 48]]])
```

```
a6[2,1::2,:]
```

```
array([[37, 38, 39, 40],
       [45, 46, 47, 48]])
```

```
a6[2,1::2,1::2]
```

```
array([[38, 40],
       [46, 48]])
```

```
a6[2,0::3,1::2]
```

```
array([[34, 36],
       [46, 48]])
```

a6

```
array([[[ 1,  2,  3,  4],
        [ 5,  6,  7,  8],
        [ 9, 10, 11, 12],
        [13, 14, 15, 16]],

       [[17, 18, 19, 20],
        [21, 22, 23, 24],
        [25, 26, 27, 28],
        [29, 30, 31, 32]],

       [[33, 34, 35, 36],
        [37, 38, 39, 40],
        [41, 42, 43, 44],
        [45, 46, 47, 48]]])
```

a6[1,:,:]

```
array([[17, 18, 19, 20],
       [21, 22, 23, 24],
       [25, 26, 27, 28],
       [29, 30, 31, 32]])
```

a6[1:3,:,:]

```
array([[[17, 18, 19, 20],
        [21, 22, 23, 24],
        [25, 26, 27, 28],
        [29, 30, 31, 32]],

       [[33, 34, 35, 36],
        [37, 38, 39, 40],
        [41, 42, 43, 44],
        [45, 46, 47, 48]]])
```

a6[0:3,:,:]

```
array([[[ 1,  2,  3,  4],
        [ 5,  6,  7,  8],
        [ 9, 10, 11, 12],
        [13, 14, 15, 16]],

       [[17, 18, 19, 20],
        [21, 22, 23, 24],
        [25, 26, 27, 28],
        [29, 30, 31, 32]],

       [[33, 34, 35, 36],
        [37, 38, 39, 40],
        [41, 42, 43, 44],
        [45, 46, 47, 48]]])
```

a6[0:2,:,:]

```
array([[[ 1,  2,  3,  4],
        [ 5,  6,  7,  8],
        [ 9, 10, 11, 12],
        [13, 14, 15, 16]],

       [[17, 18, 19, 20],
        [21, 22, 23, 24],
```

```
[25, 26, 27, 28],  
[29, 30, 31, 32]]])
```

a6[1]

```
array([[17, 18, 19, 20],  
       [21, 22, 23, 24],  
       [25, 26, 27, 28],  
       [29, 30, 31, 32]])
```

a6[1:3]

```
array([[17, 18, 19, 20],  
       [21, 22, 23, 24],  
       [25, 26, 27, 28],  
       [29, 30, 31, 32]],  
       [[33, 34, 35, 36],  
       [37, 38, 39, 40],  
       [41, 42, 43, 44],  
       [45, 46, 47, 48]])
```

a6[0:3]

```
array([[1, 2, 3, 4],  
       [5, 6, 7, 8],  
       [9, 10, 11, 12],  
       [13, 14, 15, 16]],  
       [[17, 18, 19, 20],  
       [21, 22, 23, 24],  
       [25, 26, 27, 28],  
       [29, 30, 31, 32]],  
       [[33, 34, 35, 36],  
       [37, 38, 39, 40],  
       [41, 42, 43, 44],  
       [45, 46, 47, 48]])
```

a6[0]

```
array([[1, 2, 3, 4],  
       [5, 6, 7, 8],  
       [9, 10, 11, 12],  
       [13, 14, 15, 16]])
```

a6[2]

```
array([[33, 34, 35, 36],  
       [37, 38, 39, 40],  
       [41, 42, 43, 44],  
       [45, 46, 47, 48]])
```

a6

```
array([[1, 2, 3, 4],  
       [5, 6, 7, 8],  
       [9, 10, 11, 12],  
       [13, 14, 15, 16]],  
       [[17, 18, 19, 20],  
       [21, 22, 23, 24],
```

```
[25, 26, 27, 28],  
[29, 30, 31, 32]],  
  
[[33, 34, 35, 36],  
[37, 38, 39, 40],  
[41, 42, 43, 44],  
[45, 46, 47, 48]]])
```

```
a6[0:3,1:2,:]
```

```
array([[[ 5,  6,  7,  8]],  
       [[21, 22, 23, 24]],  
       [[37, 38, 39, 40]])])
```

```
a6[1]
```

```
array([[17, 18, 19, 20],  
       [21, 22, 23, 24],  
       [25, 26, 27, 28],  
       [29, 30, 31, 32]])
```

```
a6[1,:,1:2]
```

```
array([[18],  
       [22],  
       [26],  
       [30]])
```

```
a6[1,:,1:3]
```

```
array([[18, 19],  
       [22, 23],  
       [26, 27],  
       [30, 31]])
```

```
a6[1,:,1]
```

```
array([18, 22, 26, 30])
```

```
a6[1,:,2]
```

```
array([19, 23, 27, 31])
```

```
a6
```

```
array([[[ 1,  2,  3,  4],  
        [ 5,  6,  7,  8],  
        [ 9, 10, 11, 12],  
        [13, 14, 15, 16]],  
       [[17, 18, 19, 20],  
        [21, 22, 23, 24],  
        [25, 26, 27, 28],  
        [29, 30, 31, 32]],  
       [[33, 34, 35, 36],  
        [37, 38, 39, 40],  
        [41, 42, 43, 44],  
        [45, 46, 47, 48]])])
```

```
a6[:,0,0::3]
```

```
array([[ 1,  4],  
       [33, 36]])
```

```
a6[:,0,0::3]
```

```
array([[ 1,  4],  
       [17, 20],  
       [33, 36]])
```

```
a6
```

```
array([[[ 1,  2,  3,  4],  
        [ 5,  6,  7,  8],  
        [ 9, 10, 11, 12],  
        [13, 14, 15, 16]],  
  
       [[17, 18, 19, 20],  
        [21, 22, 23, 24],  
        [25, 26, 27, 28],  
        [29, 30, 31, 32]],  
  
       [[33, 34, 35, 36],  
        [37, 38, 39, 40],  
        [41, 42, 43, 44],  
        [45, 46, 47, 48]]])
```

```
a6[:,2,1::,:]
```

```
array([[[ 5,  6,  7,  8],  
        [ 9, 10, 11, 12],  
        [13, 14, 15, 16]],  
  
       [[37, 38, 39, 40],  
        [41, 42, 43, 44],  
        [45, 46, 47, 48]]])
```

```
a6[:,2,1:3,1:]
```

```
array([[[ 6,  7,  8],  
        [10, 11, 12]],  
  
       [[38, 39, 40],  
        [42, 43, 44]]])
```

```
a6[:,2,1:3,1:4]
```

```
array([[[ 6,  7,  8],  
        [10, 11, 12]],  
  
       [[38, 39, 40],  
        [42, 43, 44]]])
```

```
a6
```

```
array([[[ 1,  2,  3,  4],  
        [ 5,  6,  7,  8],  
        [ 9, 10, 11, 12],  
        [13, 14, 15, 16]],  
  
       [[17, 18, 19, 20],
```

```
[21, 22, 23, 24],
[25, 26, 27, 28],
[29, 30, 31, 32]],

[[33, 34, 35, 36],
[37, 38, 39, 40],
[41, 42, 43, 44],
[45, 46, 47, 48]]])
```

a6[1:,]

```
array([[17, 18, 19, 20],
       [21, 22, 23, 24],
       [25, 26, 27, 28],
       [29, 30, 31, 32]],

      [[33, 34, 35, 36],
       [37, 38, 39, 40],
       [41, 42, 43, 44],
       [45, 46, 47, 48]])
```

a6[1:,0::2]

```
array([[17, 18, 19, 20],
       [25, 26, 27, 28]],

      [[33, 34, 35, 36],
       [41, 42, 43, 44]])
```

a6[1:,0::2,:]

```
array([[17, 18, 19, 20],
       [25, 26, 27, 28]],

      [[33, 34, 35, 36],
       [41, 42, 43, 44]])
```

a6[1:,0::2,1:]

```
array([[18, 19, 20],
       [26, 27, 28]],

      [[34, 35, 36],
       [42, 43, 44]])
```

a6[1:,0::2,1:3]

```
array([[18, 19],
       [26, 27]],

      [[34, 35],
       [42, 43]])
```

a6

```
array([[ 1,  2,  3,  4],
       [ 5,  6,  7,  8],
       [ 9, 10, 11, 12],
       [13, 14, 15, 16]],

      [[17, 18, 19, 20],
       [21, 22, 23, 24],
```

```
[25, 26, 27, 28],  
[29, 30, 31, 32]],  
  
[[33, 34, 35, 36],  
[37, 38, 39, 40],  
[41, 42, 43, 44],  
[45, 46, 47, 48]]])
```

```
for i in a6:  
    print(i)
```

```
[[ 1  2  3  4]  
[ 5  6  7  8]  
[ 9 10 11 12]  
[13 14 15 16]]  
[[17 18 19 20]  
[21 22 23 24]  
[25 26 27 28]  
[29 30 31 32]]  
[[33 34 35 36]  
[37 38 39 40]  
[41 42 43 44]  
[45 46 47 48]]
```

```
for i in np.nditer(a6):  
    print(i)
```

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
18  
19  
20  
21  
22  
23  
24  
25  
26  
27  
28  
29  
30  
31  
32  
33  
34  
35  
36  
37  
38
```

```
39
40
41
42
43
44
45
46
47
48
```

a6

```
array([[ 1,  2,  3,  4],
       [ 5,  6,  7,  8],
       [ 9, 10, 11, 12],
       [13, 14, 15, 16]],

      [[17, 18, 19, 20],
       [21, 22, 23, 24],
       [25, 26, 27, 28],
       [29, 30, 31, 32]],

      [[33, 34, 35, 36],
       [37, 38, 39, 40],
       [41, 42, 43, 44],
       [45, 46, 47, 48]]])
```

np.transpose(a6)

```
array([[ 1, 17, 33],
       [ 5, 21, 37],
       [ 9, 25, 41],
       [13, 29, 45]],

      [[ 2, 18, 34],
       [ 6, 22, 38],
       [10, 26, 42],
       [14, 30, 46]],

      [[ 3, 19, 35],
       [ 7, 23, 39],
       [11, 27, 43],
       [15, 31, 47]],

      [[ 4, 20, 36],
       [ 8, 24, 40],
       [12, 28, 44],
       [16, 32, 48]]])
```

a2

```
array([[ 0,  1,  2,  3],
       [ 4,  5,  6,  7],
       [ 8,  9, 10, 11]])
```

np.transpose(a2)

```
array([[ 0,  4,  8],
       [ 1,  5,  9],
       [ 2,  6, 10],
       [ 3,  7, 11]])
```



```
np.transpose(a2)
```

```
array([[ 0,  4,  8],  
       [ 1,  5,  9],  
       [ 2,  6, 10],  
       [ 3,  7, 11]])
```

```
a2.T
```

```
array([[ 0,  4,  8],  
       [ 1,  5,  9],  
       [ 2,  6, 10],  
       [ 3,  7, 11]])
```

```
a6
```

```
array([[[ 1,  2,  3,  4],  
        [ 5,  6,  7,  8],  
        [ 9, 10, 11, 12],  
        [13, 14, 15, 16]],  
       [[17, 18, 19, 20],  
        [21, 22, 23, 24],  
        [25, 26, 27, 28],  
        [29, 30, 31, 32]],  
       [[33, 34, 35, 36],  
        [37, 38, 39, 40],  
        [41, 42, 43, 44],  
        [45, 46, 47, 48]]])
```

```
a6.ravel()
```

```
array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16, 17,  
       18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34,  
       35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48])
```

```
a2
```

```
array([[ 0,  1,  2,  3],  
       [ 4,  5,  6,  7],  
       [ 8,  9, 10, 11]])
```

```
a2.ravel()
```

```
array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11])
```

```
np.arange(1,15).reshape(7,2)
```

```
array([[ 1,  2],  
       [ 3,  4],  
       [ 5,  6],  
       [ 7,  8],  
       [ 9, 10],  
       [11, 12],  
       [13, 14]])
```

```
np.arange(1,15).reshape(7,2).T
```

```
array([[ 1,  3,  5,  7,  9, 11, 13],  
       [ 2,  4,  6,  8, 10, 12, 14]])
```

```
np.arange(1,15).reshape(7,2).ravel()
```

```
array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14])
```

```
np.arange(1,13).reshape(4,3).ravel()
```

```
array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12])
```

```
np.arange(1,13).reshape(4,3).T
```

```
array([[ 1,  4,  7, 10],
       [ 2,  5,  8, 11],
       [ 3,  6,  9, 12]])
```

```
a7 = np.arange(1,10).reshape(3,3)
a8 = np.arange(10,19).reshape(3,3)
```

```
np.vstack((a7,a8))
```

```
array([[ 1,  2,  3],
       [ 4,  5,  6],
       [ 7,  8,  9],
       [10, 11, 12],
       [13, 14, 15],
       [16, 17, 18]])
```

```
np.hstack((a7,a8))
```

```
array([[ 1,  2,  3, 10, 11, 12],
       [ 4,  5,  6, 13, 14, 15],
       [ 7,  8,  9, 16, 17, 18]])
```

```
a7 = np.arange(1,10).reshape(3,3)
a8 = np.arange(10,19).reshape(3,3)
a9 = np.arange(19,28).reshape(3,3)
```

```
g = np.vstack((a7,a8,a9))
g
```

```
array([[ 1,  2,  3],
       [ 4,  5,  6],
       [ 7,  8,  9],
       [10, 11, 12],
       [13, 14, 15],
       [16, 17, 18],
       [19, 20, 21],
       [22, 23, 24],
       [25, 26, 27]])
```

```
np.hstack((a7,a8,a9))
```

```
array([[ 1,  2,  3, 10, 11, 12, 19, 20, 21],
       [ 4,  5,  6, 13, 14, 15, 22, 23, 24],
       [ 7,  8,  9, 16, 17, 18, 25, 26, 27]])
```

```
import numpy as np
```

```
df1 = np.array([[1,2,3,5],[2,3,45,6]],
               [[2,3,4,6],[8,9,5,6]])
```

```
print(df1)
```

```
[[[ 1  2  3  5]
   [ 2  3 45  6]]

 [[ 2  3  4  6]
   [ 8  9  5  6]]]
```

```
df1[1,0:,1:3]
```

```
array([[3, 4],
       [9, 5]])
```

```
import time
```

```
a11 = [i for i in range(10000000)]
b12 = [i for i in range(10000000,20000000)]
```

```
c = []
start = time.time()
for i in range(len(a)):
    c.append(a11[i] + b12[i])
print(time.time()-start)
```

```
0.0002205371856689453
```

```
import numpy as np
import time
```

```
a13 = np.arange(10000000)
b13 = np.arange(10000000,20000000)
```

```
start = time.time()
```

```
c13 = a13 + b13
print(c13)
```

```
print("Time taken:", time.time() - start)
```

```
[10000000 10000002 10000004 ... 29999994 29999996 29999998]
Time taken: 0.035874366760253906
```

```
1.1253936290740967/0.016484737396240234
```

```
68.2688235804576
```

```
# memory store
```

```
a = [i for i in range(10000000)]
import sys
sys.getsizeof(a)
```

```
89095160
```

```
b = np.arange(10000000)
sys.getsizeof(b)
```

```
80000112
```

```
b = np.arange(10000000, dtype = np.int32)
sys.getsizeof(b)
```

```
40000112
```

```
b = np.arange(10000000, dtype = np.int16)
sys.getsizeof(b)
```

```
20000112
```

```
b = np.arange(10000000, dtype = np.int8)
sys.getsizeof(b)
```

```
10000112
```

```
c = [i for i in range(100000)]
sys.getsizeof(c)
```

```
800984
```

```
d = np.arange(100000)
sys.getsizeof(d)
```

```
800112
```

```
d = np.arange(100000, dtype = np.int32)
sys.getsizeof(d)
```

```
400112
```

```
# fancy indexing
```

```
a = np.arange(24).reshape(6,4)
a
```

```
array([[ 0,  1,  2,  3],
       [ 4,  5,  6,  7],
       [ 8,  9, 10, 11],
       [12, 13, 14, 15],
       [16, 17, 18, 19],
       [20, 21, 22, 23]])
```

```
a[[0,2,3,5]]
```

```
array([[ 0,  1,  2,  3],
       [ 8,  9, 10, 11],
       [12, 13, 14, 15],
       [20, 21, 22, 23]])
```

```
a[[1,5]]
```

```
array([[ 4,  5,  6,  7],
       [20, 21, 22, 23]])
```

```
# boolean indexing
```

```
a = np.random.randint(1,100,24).reshape(6,4)
a
```

```
array([[90, 81, 74, 25],
       [61, 29, 26, 82],
       [97, 15, 85,  1],
       [33, 81, 27, 48],
       [77, 20,  1, 99],
       [ 1,  1, 39, 45]])
```

```
# find all number greater than 50
a>50
```

```
array([[ True,  True,  True, False],
       [ True, False, False,  True],
       [ True, False,  True, False],
       [False,  True, False, False],
       [ True, False, False,  True],
       [False, False, False, False]])
```

```
a[a>50]
```

```
array([90, 81, 74, 61, 82, 97, 85, 81, 77, 99])
```

```
a == 56
```

```
array([[False, False, False, False],
       [False, False, False, False],
       [False, False, False, False],
       [False, False, False, False],
       [False, False, False, False],
       [False, False, False, False]])
```

```
a[a == 56]
```

```
array([], dtype=int64)
```

```
# find out even number
```

```
a%2==0
```

```
array([[ True, False,  True, False],
       [False, False,  True,  True],
       [False, False, False, False],
       [False, False, False,  True],
       [False,  True, False, False],
       [False, False, False, False]])
```

```
a[a%2==0]
```

```
array([90, 74, 26, 82, 48, 20])
```

```
## find out odd number
```

```
a%2 !=0
```

```
array([[False,  True, False,  True],
       [ True,  True, False, False],
       [ True,  True,  True,  True],
       [ True,  True,  True, False],
```

```
[ True, False, True, True],  
[ True, True, True, True]])
```

```
a[a%2 !=0]
```

```
array([81, 25, 61, 29, 97, 15, 85, 1, 33, 81, 27, 77, 1, 99, 1, 1, 39,  
45])
```

```
# find all number greater than 50 and even number
```

```
(a>50) & (a%2==0)
```

```
array([[ True, False, True, False],  
[False, False, False, True],  
[False, False, False, False],  
[False, False, False, False],  
[False, False, False, False],  
[False, False, False, False],  
[False, False, False, False]])
```

```
a[(a>50) & (a%2==0)]
```

```
array([90, 74, 82])
```

```
# find all number greater than 50 and odd number
```

```
(a>40) & (a%2!=0)
```

```
array([[False, True, False, False],  
[ True, False, False, False],  
[ True, False, True, False],  
[False, True, False, False],  
[ True, False, False, True],  
[False, False, False, True]])
```

```
a[(a>40) & (a%2!=0)]
```

```
array([81, 61, 97, 85, 81, 77, 99, 45])
```

```
# find all number not divisible by 7
```

```
a%7 !=0
```

```
array([[ True, True, True, True],  
[ True, True, True, True],  
[ True, True, True, True],  
[ True, True, True, True],  
[False, True, True, True],  
[ True, True, True, True]])
```

```
a[a%7 !=0]
```

```
array([90, 81, 74, 25, 61, 29, 26, 82, 97, 15, 85, 1, 33, 81, 27, 48, 20,  
1, 99, 1, 1, 39, 45])
```

```
# find all number divisible by 7
```

```
a%7 == 0
```

```
array([[False, False, False, False],  
[False, False, False, False],
```

```
[False, False, False, False],  
[False, False, False, False],  
[ True, False, False, False],  
[False, False, False, False]])
```

```
a[a%7 == 0]
```

```
array([77])
```

```
b=a  
b
```

```
array([[90, 81, 74, 25],  
       [61, 29, 26, 82],  
       [97, 15, 85,  1],  
       [33, 81, 27, 48],  
       [77, 20,  1, 99],  
       [ 1,  1, 39, 45]])
```

```
# find all number divisible by 5  
a%5==0
```

```
array([[ True, False, False,  True],  
       [False, False, False, False],  
       [False,  True,  True, False],  
       [False, False, False, False],  
       [False,  True, False, False],  
       [False, False, False,  True]])
```

```
a[a%5==0]
```

```
array([90, 25, 15, 85, 20, 45])
```

```
# find all number not divisible by 5  
a[~(a%5==0)] # not operator
```

```
array([81, 74, 61, 29, 26, 82, 97,  1, 33, 81, 27, 48, 77,  1, 99,  1,  1,  
       39])
```

```
import numpy as np
```

```
def log_loss(y_true, y_pred):  
    y_true = np.array(y_true)  
    y_pred = np.array(y_pred)  
  
    loss = -(y_true*np.log(y_pred) + (1-y_true)*np.log(1-y_pred))  
  
    return np.mean(loss)
```

```
log_loss(1,.9)
```

```
np.float64(0.10536051565782628)
```

```
#
```

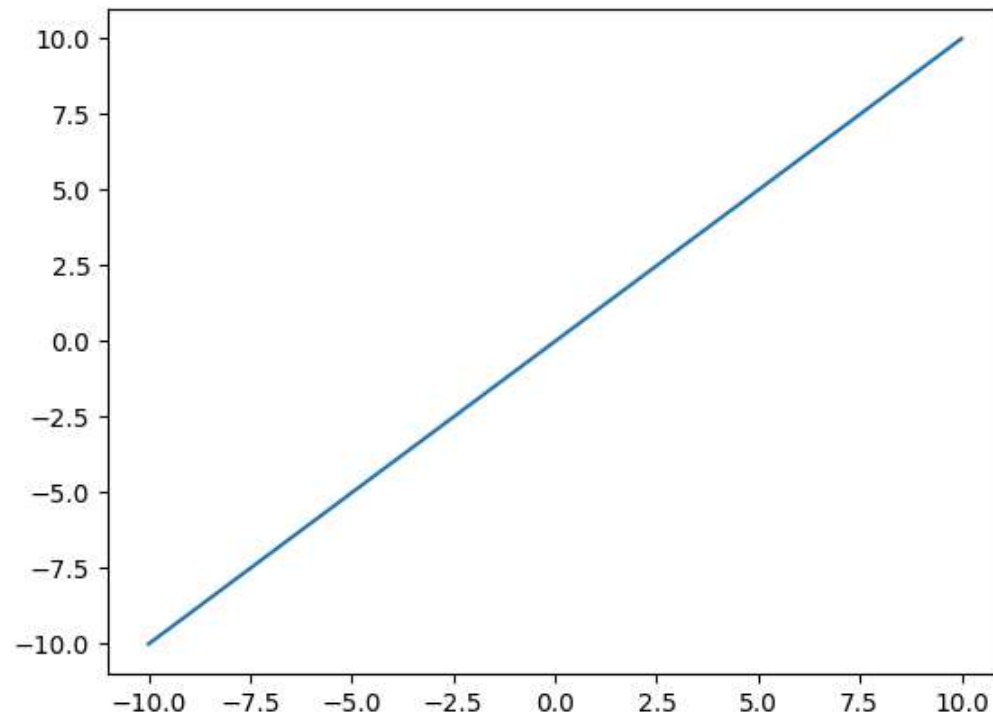
```
import matplotlib.pyplot as plt
```

```
#x = y  
x= np.linspace(-10,10,100)
```

```
y = x
```

```
plt.plot(x,y)
```

```
[<matplotlib.lines.Line2D at 0x787d4fb6ad20>]
```



```
#
```

```
import matplotlib.pyplot as plt
```

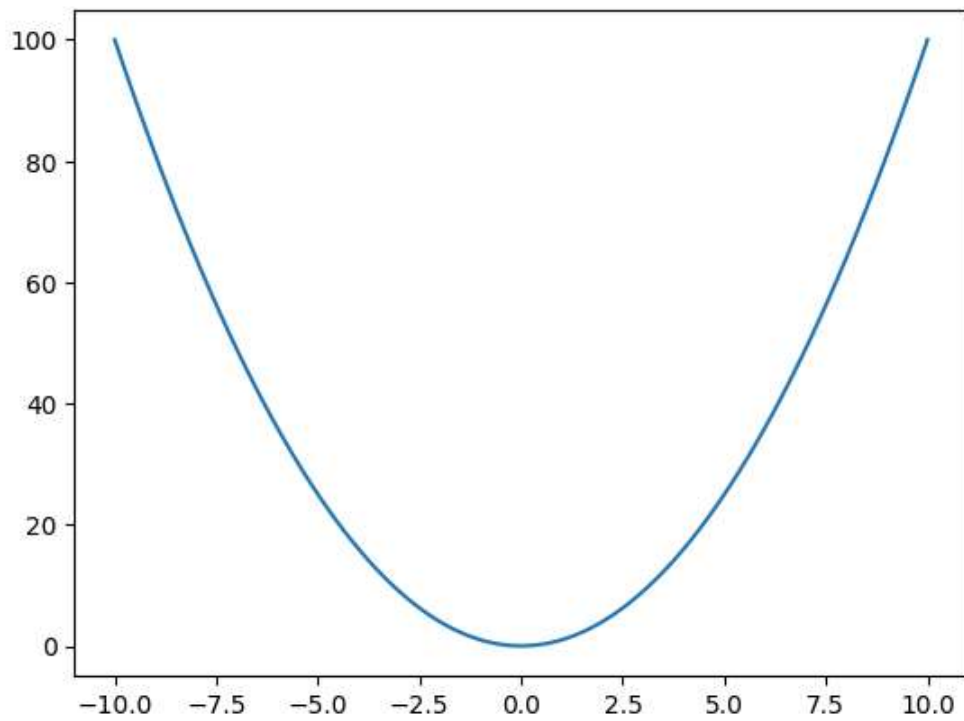
```
#y = x**2
```

```
x= np.linspace(-10,10,100)
```

```
y = x**2
```

```
plt.plot(x,y)
```

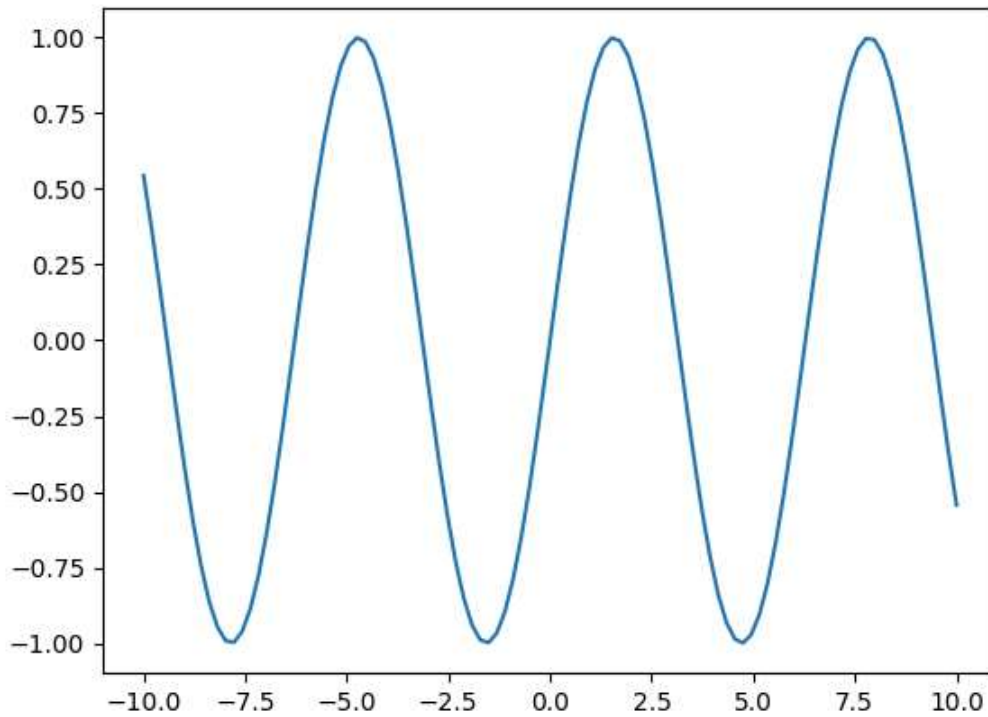

[<matplotlib.lines.Line2D at 0x787d6ee0d850>]



```
# y = sin(x)
```

```
x = np.linspace(-10,10,100)  
y = np.sin(x)  
plt.plot(x,y)
```

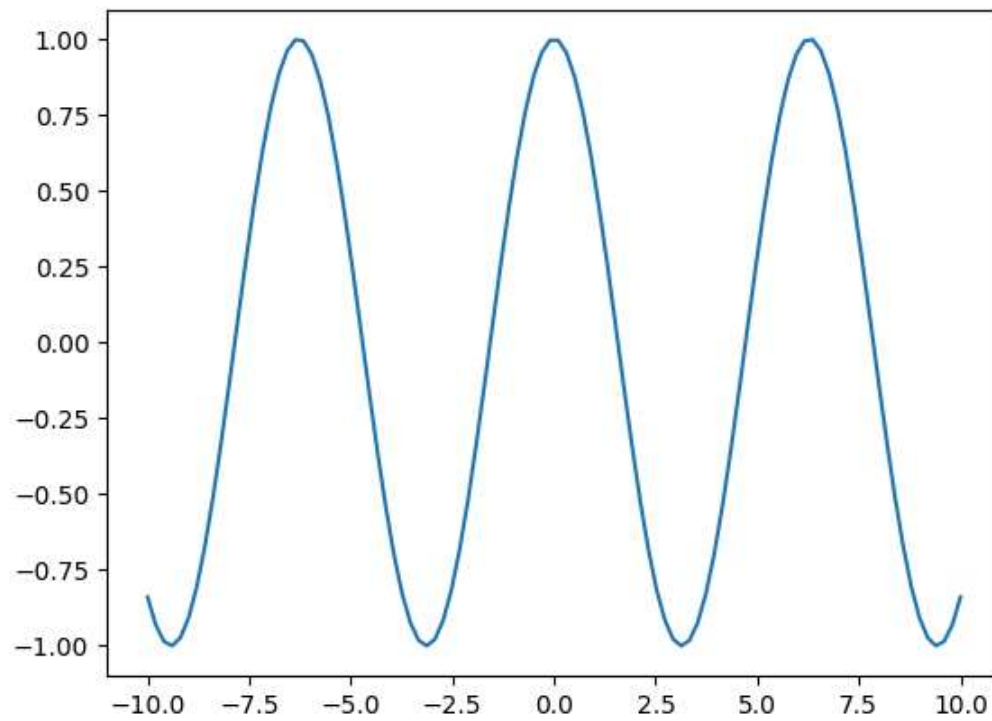
[<matplotlib.lines.Line2D at 0x787d3bdb5be0>]



```
# y = cos(x)
```

```
x = np.linspace(-10,10,100)  
y = np.cos(x)  
plt.plot(x,y)
```

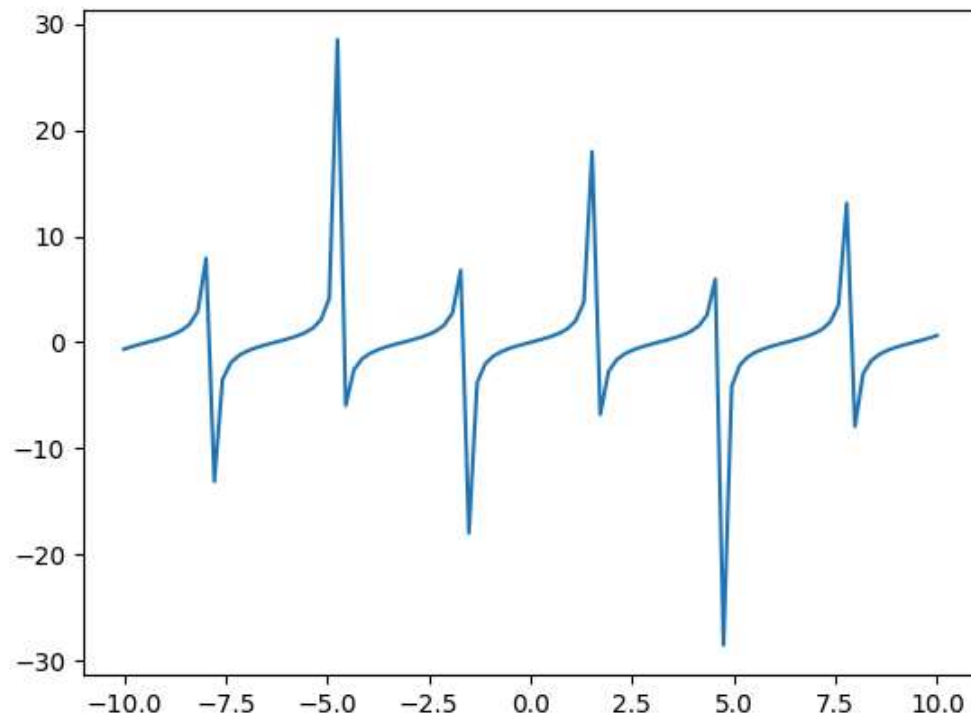
[<matplotlib.lines.Line2D at 0x787d6e9672c0>]



```
# y = tan(x)
```

```
x = np.linspace(-10,10,100)  
y = np.tan(x)  
plt.plot(x,y)
```

[<matplotlib.lines.Line2D at 0x787d290cc290>]



```
# np.sort
```

```
a = np.random.randint(1,100,15)  
a
```

```
array([17, 16,  9, 95, 86, 29, 93, 93, 17, 12, 86,  4, 39, 26, 87])
```

```
np.sort(a)
```

```
array([ 4,  9, 12, 16, 17, 17, 26, 29, 39, 86, 86, 87, 93, 93, 95])
```

```
b = np.random.randint(1,100,15).reshape(3,5)
b
```

```
array([[36,  6, 70,  9, 46],
       [46, 87, 60,  3, 18],
       [37, 62, 12, 39, 92]])
```

```
np.sort(b)
```

```
array([[ 6,  9, 36, 46, 70],
       [ 3, 18, 46, 60, 87],
       [12, 37, 39, 62, 92]])
```

```
np.sort(b,axis=0)
```

```
array([[36,  6, 12,  3, 18],
       [37, 62, 60,  9, 46],
       [46, 87, 70, 39, 92]])
```

```
#Descending Order
```

```
np.sort(b)[::-1]
```

```
array([[12, 37, 39, 62, 92],
       [ 3, 18, 46, 60, 87],
       [ 6,  9, 36, 46, 70]])
```

```
np.sort(a)[::-1]
```

```
array([95, 93, 93, 87, 86, 86, 39, 29, 26, 17, 17, 16, 12,  9,  4])
```

```
np.append(b,299)
```

```
array([ 36,  6, 70,  9, 46, 46, 87, 60,  3, 18, 37, 62, 12,
        39, 92, 299])
```

```
d2 = np.arange(12).reshape(3,4)
```

```
d3 = np.arange(12).reshape(3,4)
```

```
np.concatenate((d2,d3),axis=1)
```

```
array([[ 0,  1,  2,  3,  0,  1,  2,  3],
       [ 4,  5,  6,  7,  4,  5,  6,  7],
       [ 8,  9, 10, 11,  8,  9, 10, 11]])
```

```
d2 = np.arange(12).reshape(3,4)
```

```
d3 = np.arange(18).reshape(3,6)
```

```
d1 = np.concatenate((d2,d3),axis=1)
```

```
d1
```

```
array([[ 0,  1,  2,  3,  0,  1,  2,  3,  4,  5],
       [ 4,  5,  6,  7,  6,  7,  8,  9, 10, 11],
       [ 8,  9, 10, 11, 12, 13, 14, 15, 16, 17]])
```

```
np.unique(d1)
```

```
array([ 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16,
       17])
```

```
d1
```

```
array([[ 0, 1, 2, 3, 0, 1, 2, 3, 4, 5],
       [ 4, 5, 6, 7, 6, 7, 8, 9, 10, 11],
       [ 8, 9, 10, 11, 12, 13, 14, 15, 16, 17]])
```

```
np.sort(d1,axis=0)
```

```
array([[ 0, 1, 2, 3, 0, 1, 2, 3, 4, 5],
       [ 4, 5, 6, 7, 6, 7, 8, 9, 10, 11],
       [ 8, 9, 10, 11, 12, 13, 14, 15, 16, 17]])
```

```
d2 = np.arange(1,40)
d2
```

```
array([ 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17,
       18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34,
       35, 36, 37, 38, 39])
```

```
np.where(d2%2==0,"Even","odd")
```

```
array(['odd', 'Even', 'odd', 'Even', 'odd', 'Even', 'odd', 'Even', 'odd',
       'Even', 'odd', 'Even', 'odd', 'Even', 'odd', 'Even', 'odd', 'Even',
       'odd', 'Even', 'odd', 'Even', 'odd', 'Even', 'odd', 'Even', 'odd',
       'Even', 'odd', 'Even', 'odd', 'Even', 'odd', 'Even', 'odd', 'Even',
       'odd', 'Even', 'odd'], dtype='<U4')
```

```
np.where(d2%2==0,"Even",
        np.where(d2%2 !=0,"Even","odd"))
```

```
array(['Even', 'Even', 'Even', 'Even', 'Even', 'Even', 'Even', 'Even',
       'Even', 'Even', 'Even', 'Even', 'Even', 'Even', 'Even', 'Even',
       'Even', 'Even', 'Even', 'Even', 'Even', 'Even', 'Even', 'Even',
       'Even', 'Even', 'Even', 'Even', 'Even', 'Even', 'Even', 'Even',
       'Even', 'Even', 'Even', 'Even', 'Even', 'Even', 'Even'],
      dtype='<U4')
```

```
d2
```

```
array([ 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17,
       18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34,
       35, 36, 37, 38, 39])
```

```
np.where(d2>5,"Minor",np.where(d2>18,"Adult","old"))
```

```
array(['old', 'old', 'old', 'old', 'old', 'Minor', 'Minor', 'Minor',
       'Minor', 'Minor', 'Minor', 'Minor', 'Minor', 'Minor', 'Minor',
       'Minor', 'Minor', 'Minor', 'Minor', 'Minor', 'Minor', 'Minor',
       'Minor', 'Minor', 'Minor', 'Minor', 'Minor', 'Minor', 'Minor',
       'Minor', 'Minor', 'Minor'], dtype='<U5')
```

```
a
```

```
array([17, 16, 9, 95, 86, 29, 93, 93, 17, 12, 86, 4, 39, 26, 87])
```

```
np.where(a>50,0,a)
```

```
array([17, 16, 9, 0, 0, 29, 0, 0, 17, 12, 0, 4, 39, 26, 0])
```

```
np.where(a>50)
```

```
(array([ 3, 4, 6, 7, 10, 14]),)
```

```
np.argmax(a)
```

```
np.int64(3)
```

```
np.argmin(a)
```

```
np.int64(11)
```

```
g
```

```
array([[ 1,  2,  3],  
       [ 4,  5,  6],  
       [ 7,  8,  9],  
       [10, 11, 12],  
       [13, 14, 15],  
       [16, 17, 18],  
       [19, 20, 21],  
       [22, 23, 24],  
       [25, 26, 27]])
```

```
np.argmin(g,axis = 0)
```

```
array([0, 0, 0])
```

```
np.argmax(g,axis=1)
```

```
array([2, 2, 2, 2, 2, 2, 2, 2, 2])
```

```
a
```

```
array([17, 16, 9, 95, 86, 29, 93, 93, 17, 12, 86, 4, 39, 26, 87])
```

```
np.cumsum(a)
```

```
array([ 17,  33,  42, 137, 223, 252, 345, 438, 455, 467, 553, 557, 596,  
        622, 709])
```

```
np.cumprod(a)
```

```
array([          17,          272,          2448,  
        232560,        20000160,        580004640,  
        53940431520,        5016460131360,        85279822233120,  
        1023357866797440,        88008776544579840,        352035106178319360,  
        -4717374932755096576,        6475460264334350336,       -8484023287907620864])
```

```
g
```

```
array([[ 1,  2,  3],
       [ 4,  5,  6],
       [ 7,  8,  9],
       [10, 11, 12],
       [13, 14, 15],
       [16, 17, 18],
       [19, 20, 21],
       [22, 23, 24],
       [25, 26, 27]])
```

```
np.cumsum(g,axis=1)
```

```
array([[ 1,  3,  6],
       [ 4,  9, 15],
       [ 7, 15, 24],
       [10, 21, 33],
       [13, 27, 42],
       [16, 33, 51],
       [19, 39, 60],
       [22, 45, 69],
       [25, 51, 78]])
```

```
np.cumsum(g,axis=0)
```

```
array([[ 1,  2,  3],
       [ 5,  7,  9],
       [12, 15, 18],
       [22, 26, 30],
       [35, 40, 45],
       [51, 57, 63],
       [70, 77, 84],
       [92, 100, 108],
       [117, 126, 135]])
```

```
np.cumprod(g,axis=1)
```

```
array([[ 1,  2,  6],
       [ 4, 20, 120],
       [ 7, 56, 504],
       [10, 110, 1320],
       [13, 182, 2730],
       [16, 272, 4896],
       [19, 380, 7980],
       [22, 506, 12144],
       [25, 650, 17550]])
```

```
np.percentile(a,100)
```

```
np.float64(95.0)
```

```
np.percentile(a,97)
```

```
np.float64(94.16)
```

```
df1 = pd.read_csv("/content/credit_card.csv",sep="")
df1 = df1.squeeze()

print(type(df1))
```

```
-----  
ValueError                                Traceback (most recent call last)  
/tmp/ipykernel_390/2561352577.py in <cell line: 0>()  
----> 1 df1 = pd.read_csv("/content/credit_card.csv", sep="")  
      2 df1 = df1.squeeze()  
      3  
      4 print(type(df1))
```

```
----- 4 frames -----  
/usr/local/lib/python3.12/dist-packages/pandas/io/parsers/c_parser_wrapper.py in __init__(self, src,   
**kwds)  
      91         # Fail here loudly instead of in cython after reading  
      92         import_optional_dependency("pyarrow")  
----> 93         self._reader = parsers.TextReader(src, **kwds)  
      94  
      95         self.unnamed_cols = self._reader.unnamed_cols  
  
parsers.pyx in pandas._libs.parsers.TextReader.__cinit__()
```

ValueError: only single character unicode strings can be converted to Py_UCS4, got length 0
Next steps: [Explain error](#)

```
df1.describe()
```

```
df1 = pd.read_csv("/content/credit_card.csv")  
series = df1.squeeze()  
  
print(type(series))
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
import pandas as pd  
  
df1 = pd.read_csv("/content/credit_card.csv", index_col="Client_Num")  
df1 = df1.squeeze()  
  
print(type(df1))
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
df1.head()
```

	Card_Category	Annual_Fees	Activation_30_Days	Customer_Acq_Cost	Week_Start_Date	Week_N
Client_Num						
708082083	Blue	200	0	87	01-01-2023	Week
708083283	Blue	445	1	108	01-01-2023	Week
708084558	Blue	140	0	106	01-01-2023	Week
708085458	Blue	250	1	150	01-01-2023	Week
708086958	Blue	320	1	106	01-01-2023	Week